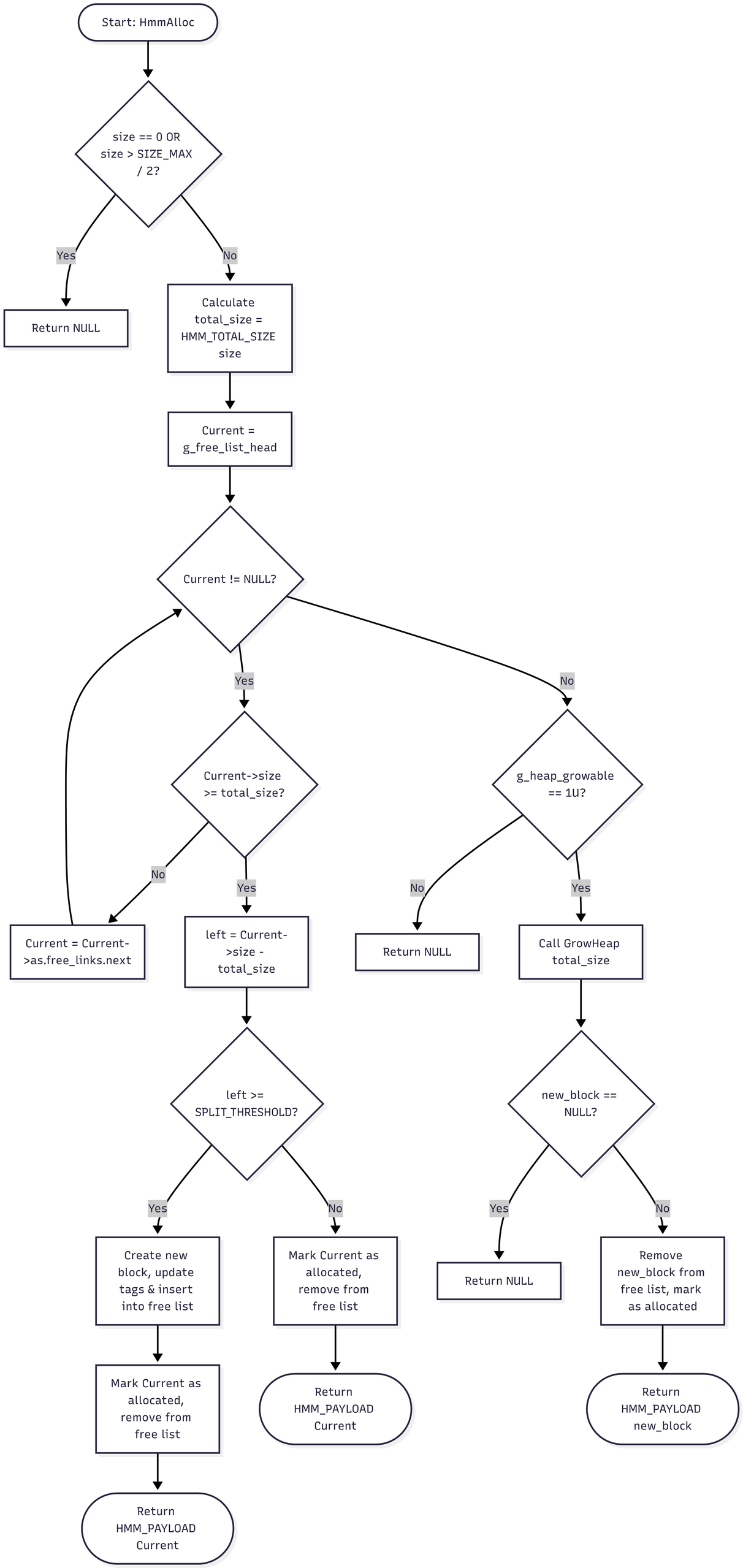

Start: HmmAlloc
size == 0 OR size > SIZE_MAX / 2?
Yes
No
Return NULL
Calculate total_size = HMM_TOTAL_SIZE size
Current = g_free_list_head
Current != NULL?
Yes
No
Current->size >= total_size?
No
Yes
Current = Current->as.free_links.next
left = Current->size - total_size
left >= SPLIT_THRESHOLD?
Yes
No
Create new block, update tags & insert into free list
Mark Current as allocated, remove from free list
Mark Current as allocated, remove from free list
Return HMM_PAYLOAD Current
Return HMM_PAYLOAD Current
g_heap_growable == 1U?
No
Yes
Return NULL
Call GrowHeap total_size
new_block == NULL?
Yes
No
Return NULL
Remove new_block from free list, mark as allocated
Return HMM_PAYLOAD new_block